

13. Roman to Integer:

Roman numerals are represented by seven different symbols: **I**, **V**, **X**, **L**, **C**, **D** and **M**.

Symbol	Value
I	1
V	5
X	10
L	50
C	100
D	500
M	1000

For example, **2** is written as **II** in Roman numerals, just two ones added together. **12** is written as **XII**, which is simply **X + II**. The number **27** is written as **XXVII**, which is **XX + V + II**.

Roman numerals are usually written from largest to smallest from left to right. However, the numeral for four is not **IIII**. Instead, the number four is written as **IV**. Because the one is before the five, we subtract it, making four. The same principle applies to the number nine, which is written as **IX**. There are six instances where subtraction is used:

- **I** can be placed before **V** (5) and **X** (10) to make 4 and 9.
- **X** can be placed before **L** (50) and **C** (100) to make 40 and 90.
- **C** can be placed before **D** (500) and **M** (1000) to make 400 and 900.

Given a Roman numeral, convert it to an integer.

Example 1:

Input: `s = "III"`
Output: 3
Explanation: `III = 3`.

Example 2:

Input: s = "LVIII"

Output: 58

Explanation: L = 50, V= 5, III = 3.

Example 3:

Input: s = "MCMXCIV"

Output: 1994

Explanation: M = 1000, CM = 900, XC = 90 and IV = 4.

Constraints:

- `1 <= s.length <= 15`
- `s` contains only the characters `('I', 'V', 'X', 'L', 'C', 'D', 'M')`.
- It is **guaranteed** that `s` is a valid roman numeral in the range `[1, 3999]`.

Solution:

```
class Solution:
    def romanToInt(self, s: str) -> int:
        total = 0
        roman = { 'I':1, 'V':5, 'X':10, 'L':50, 'C':100,
                  'D':500, 'M':1000 }
        for i in range(len(s)):
            if i+1 < len(s) and roman[s[i]] < roman[s[i+1]]:
                total -= roman[s[i]]
            else:
                total += roman[s[i]]
        return total
```

alternative: reverse scan:

```
class Solution:
    def romanToInt(self, s: str) -> int:
        roman = {'I':1, 'V':5, 'X':10, 'L':50, 'C':100,
```

```
'D':500, 'M':1000}
    total = 0
    prev = 0

    for ch in reversed(s):
        if roman[ch] < prev:
            total -= roman[ch]
        else:
            total += roman[ch]
        prev = roman[ch]

    return total
```